HAMED HOJATIAN

Ph.D. Candidate

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Community Service: IEEE Journals Reviewer. Details in here.

EDUCATION

Ph.D. on Applied Deep Learning in wireless communication Systems

Polytechnique Montréal

Fall 2019 - current — Montreal, Canada

Supervisor: **Prof. François Leduc-Primeau**
Co-Supervisor: **Prof. Jean-François Frigon**
CGPA: **4.0/4.0**

M.Sc. in Telecommunication Engineering

Isfahan University of Technology (IUT)

2014 - 2017 — Isfahan, Iran

Supervisor: **Prof. Mohammad Javad Omidi**
CGPA: **3.7/4.0**

B.Sc. in Electrical Engineering

Babol (Noshirvani) University of technology (NIT)

2010 - 2014 — Mazandaran, Iran

Supervisor: **Prof. Hosein Miar-Naeimi**
CGPA: **3.1/4.0**

DISSERTATION WORK

Ph.D. Thesis

Massive MIMO Hybrid Beamforming Design with Deep Learning

- RSSI-Based hybrid beamforming design with deep learning;
- Improve the spectral and energy efficiency of hybrid beamforming design;
- Unsupervised learning for hybrid beamforming design and intelligent beamforming;
- Coordinated hybrid beamforming and cell-free massive MIMO with deep neural network; and
- ...

M.Sc. Thesis

Time and Frequency synchronization in Massive MIMO systems

- Carrier Frequency Offset (CFO) and channel estimation Using ZC sequence;
- Detection in receiver using iterative method; and
- Improve the method using Filtered- OFDM.

B.Sc. Thesis

OFDM Transmitter and receiver

- Developed in MATLAB simulation
- OFDMA multi-user simulation in AWGN channel.

SEEKING FOR

"To work in a progressive and dynamic research organization where one could solve challenging scientific problems"

IT SKILLS

Python	●●●●○
Pytorch	●●●●●
MATLAB	●●●○○
LaTeX	●●●●○
MS Office	●●●●○
HTML5 and CSS	●●○○○
JAVA SCRIPT	●●○○○
Visual basic	●●○○○
Bash	●●○○○

LANGUAGES

English	●●●●○
French	●○○○○

TELECOMM. TOOLS

Ericsson OSS	●●●●○
MapInfo	●●●●○
Business Object (BO)	●●●○○
Atoll	●●●○○
TEMS	●●○○○
Schema	●●○○○
Actix	●●○○○

COURSES

Polytechnique Montréal (PolyMTL)

- Learning Representation (A+)
  - instructor: Prof. Aaron Courville
- Wireless Communication (A)
  - instructor: Prof. Jean-François Frigon
- Estimation and Detection Theory (A)
  - instructor: Prof. Jean-François Frigon

Isfahan University of Technology (IUT)

- Advance Digital Communication (ADC)
- Information Theory
- Software Define Radio (SDR)
- Digital Signal Processing (DSP)

RESEARCH INTERESTS

- Applied ML in wireless communication;
- Machine Learning, Deep Learning, DRL;
- Optimization;
- Signal processing in Massive MIMO communication;
- Millimeter wave in Massive MIMO and Antenna array;
- Coordinated, cooperative beamforming and distributed antenna; and
- Cell free communication.

PUBLICATION

- H. Hojatian, J. Nadal, J.-F. Frigon, and F. Leduc-Primeau.
  "Decentralized Beamforming for Cell-Free Massive MIMO with Unsupervised Learning",
  IEEE Communications Letters, 2022.
- H. Hojatian, J. Nadal, J.-F. Frigon, and F. Leduc-Primeau.
  "Unsupervised Deep Learning for Massive MIMO Hybrid Beamforming",
  IEEE Transactions on Wireless Communications, 2021.
- H. Hojatian, V. N. Ha, J. Nadal, J.-F. Frigon, and F. Leduc-Primeau.
  "RSSI-Based Hybrid Beamforming Design with Deep Learning",
  IEEE International Conference on Communications (ICC), 2020.
- H. Hojatian, M. J. Omidi, H. Saeedi-Sourk, A. Farhang,
  "Joint CFO and channel estimation in OFDM-based massive MIMO systems",
  8th International Symposium on Telecommunications (IST), 2016.

WORK EXPERIENCE

Senior RAN (2G, 3G and LTE) Engineer

Mobile Telecommunication Company of Iran (MCI)

Nov 2015 - Nov 2018 | Tehran, Iran

- Ericsson OSS in Tehran Network;
- Network Planning (2G, 3G and LTE);
- Network Optimization (BSC owner and RNC owner);
- Project manager of RNC rehoming project in Tehran; and
- Parameter tuning and optimization.

Vice-President

Eleventh (11th) national robotics competition at BABOL University of Technology

Oct 2015 | Mazandaran, Iran

CERTIFICATES

- "**Introduction to Unix Shell**" *Calcul Quebec*, 2022, Montreal, Canada.
- "**Massive data analysis with Spark**" *Calcul Quebec*, 2022, Montreal, Canada.
- "**Introduction to GPU Programming with Python**" *Calcul Quebec*, 2022, Montreal, Canada.
- "**4TH IVADO/MILA SCHOOL IN DEEP LEARNING**" *IVADO/MILA*, 2019, Montreal, Canada.
- "**DEEP LEARNING WITH PYTORCH**" *Udemy*, 2018, Online.
- "**LTE OPTIMIZATION OF RADIO NETWROK (3GPP)**" *TECHCOM Consulting GmbH*, 2018, Tehran, Iran.
- "**LTE COUNTER AND KPI (ERICSSON)**" *TECHCOM Consulting GmbH*, 2017, Tehran, Iran.
- "**LTE RADIO NETWORK PLANNING**" *TECHCOM Consulting GmbH*, 2017, Tehran, Iran.

TALKS & TEACHING

Author

IEEE ICC2020

Jun 2020 | Virtual-Dublin, Ireland

- Present my accepted paper in ICC2020;

Teaching assistant

BNUT

2013 | Mazandaran, Iran

- Teaching assistant for Electronic I
- Teaching assistant for Electronic circuit I
- Teaching assistant for Digital Signal Processing

STRENGTHS

Hard-working | Decision Making | Ability to Work Under Pressure | Motivator & Leader | Conflict Resolution | Time Management | Target oriented | Flexibility and Adaptability

HOBBIES

Music
Listen quality music of every genre

Searching
Always eager to find the updated topics in my fields

Swimming
Fresh water is my charger